

## HOHE WAND HOCHKOGELHAUS, Austria

WMO: 113850

Lat: 47.822N

Lon: 16.035E

Elev: 3075

StdP: 13.13

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 10.2       | 13.9 | 3.4                           | 7.3 | 15.9 | 8.0 | 9.3 | 19.2 | 26.2                  | 17.4 | 23.3 | 19.9 | 10.2                  | 300  | 0.585 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 13.6                   | 81.2            | 66.0 | 77.5 | 64.6 | 74.5 | 63.0 | 67.8                | 77.8 | 65.8 | 75.3 | 64.0 | 72.9 | 8.3                  | 120  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.1                            | 100.6 | 74.2 | 62.2 | 93.7 | 71.6 | 60.3 | 87.7 | 69.4 | 34.1          | 78.1 | 32.4 | 75.5 | 30.9 | 73.1 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 23.1          | 20.3 | 17.8 | DB                         | 7.9 | 86.7               | 4.7 | 4.5                                                | 90.0 | 1.8        | 92.6 | -0.9       | 95.2 | -4.3       | 98.5 |  |  |
| (5)               |      |      | WB                         | 6.5 | 71.4               | 4.6 | 2.2                                                | 73.0 | 0.5        | 74.2 | -2.1       | 75.4 | -5.5       | 77.0 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.3   | 28.9  | 31.1  | 36.3  | 45.8  | 53.0  | 59.5  | 62.8  | 63.2  | 54.4  | 47.5  | 39.4  | 32.4 |
|                                                                                  | DBStd       | 14.53  | 8.88  | 9.57  | 8.92  | 8.90  | 7.65  | 7.64  | 7.01  | 7.09  | 7.53  | 8.52  | 8.48  | 8.70 |
|                                                                                  | HDD50       | 2929   | 653   | 530   | 429   | 179   | 57    | 11    | 1     | 2     | 39    | 151   | 331   | 546  |
|                                                                                  | HDD65       | 7002   | 1118  | 949   | 888   | 577   | 377   | 199   | 127   | 120   | 325   | 543   | 769   | 1010 |
|                                                                                  | CDD50       | 1571   | 0     | 1     | 6     | 52    | 150   | 295   | 398   | 410   | 172   | 74    | 12    | 1    |
|                                                                                  | CDD65       | 167    | 0     | 0     | 0     | 0     | 5     | 32    | 59    | 63    | 8     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 796    | 0     | 0     | 0     | 0     | 13    | 135   | 304   | 322   | 22    | 0     | 0     | 0    |
|                                                                                  | CDH80       | 152    | 0     | 0     | 0     | 0     | 1     | 18    | 59    | 71    | 3     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.6    | 7.5   | 7.9   | 7.5   | 8.1   | 8.6   | 7.6   | 8.0   | 7.7   | 7.5   | 6.9   | 7.0   | 6.9  |
| Precipitation                                                                    | PrecAvg     | 30.5   | 1.6   | 1.5   | 2.0   | 2.0   | 3.3   | 4.0   | 3.7   | 3.5   | 2.7   | 2.2   | 2.1   | 1.8  |
|                                                                                  | PrecMax     | 38.9   | 4.4   | 3.8   | 5.0   | 4.4   | 7.0   | 8.0   | 8.5   | 6.5   | 7.7   | 3.9   | 4.7   | 3.9  |
|                                                                                  | PrecMin     | 20.7   | 0.2   | 0.1   | 0.6   | 0.3   | 0.7   | 1.2   | 0.4   | 0.9   | 0.4   | 0.4   | 0.0   | 0.2  |
|                                                                                  | PrecStd     | 4.6    | 1.0   | 1.0   | 1.3   | 1.0   | 1.6   | 1.8   | 1.9   | 1.4   | 1.6   | 1.1   | 0.9   | 0.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 53.0  | 55.6  | 60.5  | 70.7  | 75.5  | 82.5  | 85.5  | 86.4  | 76.6  | 69.8  | 61.0  | 54.2 |
|                                                                                  |             | MCWB   | 43.0  | 43.6  | 48.6  | 56.0  | 61.6  | 67.1  | 67.5  | 65.9  | 63.8  | 56.7  | 49.8  | 43.2 |
|                                                                                  | 2%          | DB     | 48.5  | 50.7  | 56.1  | 65.5  | 71.6  | 77.8  | 80.9  | 81.4  | 71.9  | 65.4  | 57.1  | 49.9 |
|                                                                                  |             | MCWB   | 40.6  | 40.7  | 45.8  | 53.3  | 59.9  | 65.3  | 66.5  | 65.6  | 61.3  | 54.9  | 48.4  | 41.3 |
|                                                                                  | 5%          | DB     | 45.1  | 47.3  | 51.9  | 61.8  | 68.0  | 74.4  | 77.0  | 77.2  | 67.8  | 61.8  | 54.0  | 47.0 |
|                                                                                  |             | MCWB   | 39.0  | 39.5  | 43.5  | 51.0  | 57.9  | 63.3  | 64.6  | 64.1  | 59.1  | 53.4  | 47.1  | 40.1 |
|                                                                                  | 10%         | DB     | 41.6  | 43.8  | 48.3  | 58.1  | 64.7  | 71.1  | 73.7  | 73.9  | 64.4  | 58.6  | 51.0  | 44.4 |
|                                                                                  |             | MCWB   | 37.0  | 37.8  | 41.7  | 48.8  | 55.7  | 61.4  | 63.1  | 62.7  | 57.3  | 51.9  | 45.3  | 39.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.6  | 45.0  | 49.5  | 57.0  | 64.2  | 69.7  | 70.9  | 70.0  | 64.9  | 58.9  | 52.1  | 45.2 |
|                                                                                  |             | MCDB   | 49.6  | 52.6  | 59.0  | 68.8  | 73.2  | 78.2  | 81.6  | 81.7  | 73.8  | 66.9  | 57.7  | 50.8 |
|                                                                                  | 2%          | WB     | 41.8  | 42.4  | 46.6  | 54.1  | 61.0  | 66.6  | 68.1  | 67.1  | 62.2  | 56.4  | 49.7  | 43.2 |
|                                                                                  |             | MCDB   | 47.1  | 49.0  | 54.5  | 63.9  | 69.4  | 75.7  | 77.9  | 77.9  | 69.9  | 63.2  | 55.1  | 48.0 |
|                                                                                  | 5%          | WB     | 39.7  | 40.3  | 44.4  | 51.8  | 58.7  | 64.3  | 65.8  | 65.2  | 60.1  | 54.5  | 47.7  | 41.4 |
|                                                                                  |             | MCDB   | 44.5  | 46.0  | 50.5  | 60.3  | 66.8  | 73.1  | 74.9  | 75.2  | 66.5  | 60.4  | 52.8  | 45.8 |
|                                                                                  | 10%         | WB     | 37.3  | 38.3  | 42.5  | 49.7  | 56.3  | 62.1  | 63.7  | 63.3  | 58.1  | 52.6  | 45.8  | 39.2 |
|                                                                                  |             | MCDB   | 41.2  | 43.4  | 47.5  | 56.8  | 63.5  | 70.0  | 72.4  | 72.3  | 63.7  | 57.8  | 50.2  | 43.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.5   | 9.1   | 10.6  | 13.0  | 13.0  | 13.1  | 13.7  | 13.6  | 11.5  | 10.6  | 8.8   | 9.0  |
|                                                                                  |             | MCDBR  | 12.1  | 12.7  | 15.5  | 17.1  | 17.0  | 17.1  | 18.4  | 18.3  | 16.1  | 14.0  | 12.3  | 12.7 |
|                                                                                  |             | MCWBR  | 7.7   | 8.1   | 9.9   | 10.0  | 10.4  | 10.5  | 9.7   | 9.6   | 9.8   | 8.8   | 7.7   | 8.1  |
|                                                                                  | 5% WB       | MCDBR  | 11.0  | 11.6  | 14.4  | 15.9  | 16.5  | 15.9  | 17.0  | 17.3  | 15.3  | 13.0  | 11.1  | 11.6 |
|                                                                                  |             | MCWBR  | 8.0   | 8.2   | 9.8   | 9.5   | 10.6  | 10.6  | 10.1  | 10.1  | 9.9   | 8.6   | 7.6   | 8.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.278  | 0.300 | 0.334 | 0.371 | 0.372 | 0.378 | 0.382 | 0.381 | 0.357 | 0.331 | 0.307 | 0.278 |      |
|                                                                                  | taud        | 2.448  | 2.420 | 2.374 | 2.310 | 2.339 | 2.347 | 2.332 | 2.349 | 2.399 | 2.460 | 2.491 | 2.482 |      |
|                                                                                  | Ebn at Noon | 258    | 275   | 280   | 279   | 282   | 280   | 277   | 272   | 269   | 259   | 244   | 244   |      |
|                                                                                  | Edh at Noon | 20     | 26    | 32    | 38    | 38    | 38    | 38    | 36    | 31    | 25    | 20    | 18    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 342    | 609   | 965   | 1383  | 1622  | 1768  | 1730  | 1542  | 1107  | 710   | 378   | 272   |      |
|                                                                                  | RadStd      | 38     | 80    | 117   | 143   | 175   | 134   | 153   | 159   | 135   | 73    | 43    | 34    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (92 neighbors) | +0.99 | N/A     | N/A    | +1.49   | +0.79 | +0.72 | -211        | -270  | +151  | +81   |

Nomenclature: See separate page